

Notes: Schneider, Strahan, and Yang (JFE, 2025)

Bank stress testing, human capital investment and risk management

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1 Big picture

- **Question.** Do post-crisis bank stress tests change banks’ investments in risk-management human capital, and do they improve broad risk-management capacity or only test-specific capacity?
- **Core setting.** The paper studies U.S. banks from 2010 to 2021 using Lightcast job-posting data merged to bank holding companies and FR Y-9C financial data.
- **Measurement innovation.** The authors construct broad *RM Jobs* and NLP-based *RM Skills*, then decompose RM Skills into Operations, Technology, Lending, Compliance, and Capital Regulation.
- **Main finding.** Stress tests raise demand for stress-test-specific human capital, especially highly educated workers and skills related to Technology and Capital Regulation.
- **Important limitation.** Stress tests do not appear to raise broad, non-test-specific RM demand. Their effect is concentrated on jobs explicitly connected to passing or preparing for the tests.
- **Economic takeaway.** Stress tests can force banks to build specialized risk-management capacity, but the evidence is mixed on whether this improves broad risk-management culture.

2 Data and measurement

2.1 Lightcast job postings and bank matching

The paper uses Lightcast online job-posting data for 2010–2021 and links postings to banks in the top 300 by assets. Bank characteristics come from FR Y-9C reports. Job postings reveal intended human-capital demand before actual employment changes are visible.

2.2 RM Jobs and RM Skills

RM Jobs are posts requiring risk-management skills or tasks. *RM Skills* are constructed from a structural topic model over job-skill phrases and scaled by total job posts. The topic model removes generic skills and keeps risk-management-relevant topics.

2.3 Skill subcategories

RM Skills are split into Operations, Technology, Lending, Compliance, and Capital Regulation. This split lets the paper test whether regulation raises broad RM demand or mostly regulatory and stress-test-specific skill demand.

2.4 Stress-test variables

Large banks have assets of at least \$50 billion. *Stress Tested* indicates exposure to the next test. *Stress Test Jobs* are RM jobs requiring “Stress Testing” or “CCAR Regulatory Rules.” Failure and exposure variables use CCAR failure scores and capital-ratio exposure measures.

3 Models and empirical specifications

3.1 Cross-sectional RM demand model

$$RM_{i,t} = \beta_1 Volatility(ROA)_{i,t-1} + \beta_2 ROA_{i,t-1} + \beta_3 Deposits/Assets_{i,t-1} + \beta_4 CapitalRatio_{i,t-1} + \beta_5 StressTested_{i,t+1} + \beta_6 LargeBank_i + \beta_7 Complexity_{i,t-1} + Controls_{i,t} + \alpha_t + \varepsilon_{i,t}.$$

Interpretation. The dependent variable can be RM Jobs, RM Skills, or a subcategory. The model tests whether risk, performance, capital structure, size, and stress-test exposure predict RM hiring.

3.2 Panel model around stress-test exposure

$$RM_{i,t} = \alpha_t + \eta_i + \beta StressTested_{i,t+1} + ControlVariables_{i,t} + \varepsilon_{i,t}.$$

Interpretation. Bank fixed effects remove time-invariant bank characteristics. The coefficient measures whether banks hire more when they are scheduled for a stress test.

3.3 Failure and exposure model

$$RM_{i,t} = \alpha_t + \eta_i + \beta_1 CCARFailureScore_{i,t-1} + \beta_2 Tier1CapExposure_{i,t-1} + \beta_3 TRBCExposure_{i,t-1} + \beta_4 Tier1LeverageExposure_{i,t-1} + Controls_{i,t} + \varepsilon_{i,t}.$$

Interpretation. This asks whether poor stress-test performance leads banks to hire more stress-test-related RM labor.

4 Identification and empirical design

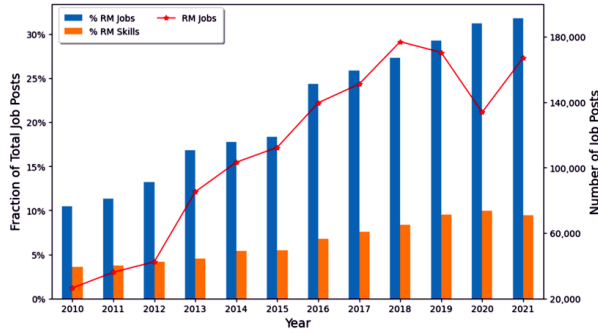
The paper is not a randomized experiment. Its credibility comes from (i) novel measurement of RM human-capital demand from job postings, (ii) within-bank panel variation in stress-test exposure, (iii) bank and year fixed effects, and (iv) dynamic difference-in-difference evidence around the initiation of testing. The key limitation is that job postings measure demand for RM labor, not realized risk-management quality.

5 Tables and figures

5.1 Figure 1: Labor Demand for RM Jobs and RM Skills; Table 2: Summary Statistics

Read: Figure 1 shows RM Jobs rising from roughly 10% of posts in 2010 to above 30% by 2020–2021, while RM Skills rise from about 3–4% to nearly 10%. Table 2 shows that RM Jobs average 20.86% of postings and RM Skills average 6.64%.

Economic message. Risk management becomes a much larger part of bank labor demand after the crisis, but precise quantitative/regulatory skills remain a smaller subset.



(a) Figure 1: Labor Demand for RM Jobs and RM Skills

Table 2
Summary statistics.

	N	Mean	Std	p25	p50	p75
Panel A: Labor Demand						
A.1 All Banks						
RM Jobs _{it}	3706	20.86	24.79	3.51	11.54	28.00
RM Jobs – Higher Edu _{it}	3706	0.32	0.82	0.00	0.00	0.18
RM Jobs – Low Edu _{it}	3706	20.48	24.65	3.28	11.11	26.78
RM Skills _{it}	3706	6.64	7.69	1.15	4.15	9.11
Operations _{it}	3706	1.26	1.42	0.21	0.79	1.82
Tech _{it}	3706	0.74	1.23	0.01	0.26	0.90
Lending _{it}	3706	2.20	2.68	0.26	1.32	3.04
Compliance _{it}	3706	1.93	3.31	0.15	0.64	1.87
Capital Regulation _{it}	3706	0.34	0.59	0.01	0.13	0.40
A.2 Small Banks (Assets < \$50 b)						
RM Jobs _{it}	3288	20.68	25.64	2.33	10.40	27.95
RM Jobs – Higher Edu _{it}	3288	0.20	0.68	0.00	0.00	0.00
RM Jobs – Low Edu _{it}	3288	20.42	25.52	2.23	10.09	26.89
RM Skills _{it}	3288	6.62	8.02	0.79	3.83	8.99
Operations _{it}	3288	1.20	1.44	0.14	0.70	1.71
Tech _{it}	3288	0.60	1.13	0.00	0.19	0.67
Lending _{it}	3288	2.31	2.81	0.14	1.37	3.33
Compliance _{it}	3288	2.08	3.49	0.10	0.66	2.19
Capital Regulation _{it}	3288	0.25	0.45	0.00	0.10	0.30
A.3 Large Banks (Assets ≥ \$50 b)						
RM Jobs _{it}	418	22.30	16.54	10.36	18.33	28.60
RM Jobs – Higher Edu _{it}	418	1.28	1.14	0.47	0.95	1.69
RM Jobs – Low Edu _{it}	418	21.00	16.25	9.67	16.59	26.19
RM Skills _{it}	418	6.86	4.17	3.66	6.08	9.38
Operations _{it}	418	1.68	1.15	0.84	1.46	2.25
Tech _{it}	418	1.82	1.42	0.72	1.43	2.62
Lending _{it}	418	1.37	0.88	0.76	1.14	1.70
Compliance _{it}	418	0.76	0.55	0.37	0.59	0.99
Capital Regulation _{it}	418	1.11	0.91	0.47	0.79	1.52
N		Mean	Std	p25	p50	p75
Panel B: Bank Characteristics						
Volatility of ROA _{it-1}	3706	0.23	0.47	0.04	0.08	0.19
ROA _{it-1}	3706	0.83	0.90	0.60	0.89	1.15
Loans&Commit/Assets _{it-1}	3706	84.37	20.73	75.07	85.54	94.80
Deposits/Assets _{it-1}	3706	76.22	12.89	73.59	79.42	83.46
Capital Ratio _{it-1}	3706	14.10	5.12	11.43	12.88	15.04
Log(Assets) _{it-1}	3706	15.67	1.60	14.56	15.27	16.38
Stress Tested _{it-1}	3706	0.08	0.28	0.00	0.00	0.00
Complexity _{it-1}	3706	2.40	1.38	1.61	2.20	2.94
Panel C: Additional Metrics for Large Banks						
Expected Shortfall _{it-1}	398	41.83	6.98	36.45	41.63	46.11
Beta _{it-1}	398	1.08	0.24	0.89	1.05	1.21
Volatility _{it-1}	398	25.76	9.23	18.55	23.44	31.01
Long-term Investors _{it-1}	418	43.12	26.56	25.98	53.23	63.50
Delta _{it-1}	418	0.64	1.47	0.00	0.06	0.48
Vega _{it-1}	418	0.16	0.34	0.00	0.00	0.13

(b) Table 2: Summary Statistics

5.2 Table 1: Hypothetical Example to Illustrate Construction of RM-Skill

Read: Table 1 shows how job descriptions and skill phrases are mapped into topic scores and then into bank-year RM Skills.

Economic message. The table makes the measurement transparent: the outcome is not a simple keyword count but a topic-model score for risk-management skill intensity.

Table 1

Hypothetical example to illustrate construction of RM-Skill.

This table includes 5 Panels designed to make transparent our construction of bank demand for risk-management skills, which is summarized in the variable RM-Skill. See Section 3 of the paper for a full discussion.

Panel 1. This exhibit provides a hypothetical example of a post that may appear in the Lightcast data. The bold text are phrases that Lightcast may identify as job skills. The skills appear next to Job #1 in Exhibit 2.

Job #1. This position requires experience developing, executing, and integrating a broad spectrum of **Basel** initiatives requiring the use of leading-edge methodologies and solutions. You will leverage your expertise in financial **economics** to analyze market trends and their impact on financial stability. This critical role involves developing and implementing complex **stress testing** frameworks and evaluating the resilience of portfolios under adverse scenarios including for **CCAR** compliance. The ideal candidate will have familiarity with **derivatives** and other financial instruments of significant technical complexity. You will **collaborate** within cross-functional **teams** of credit analysts, data scientists, and IT professionals. Effective **communication** of risk findings and recommendations to senior management and regulatory authorities is a critical aspect of the responsibilities. You will contribute to maintaining financial stability and promoting sound **risk management** practices in the bank.

Panel 2. This exhibit presents hypothetical skill lists for eight RM jobs across two banks, Bank A and Bank B, as they might appear in the Lightcast data. Job #1 represents the skills extracted from the example job post in Exhibit 1.

Job	Bank	Skills
#1	A	Basel, CCAR, Communication, Derivatives, Economics, Risk Management, Stress Testing, Teamwork
#2	A	Communication, Management, Multitasking, Presentations, Risk Management, Teamwork, Troubleshooting, Writing
#3	A	CCAR, Economics, Multitasking, Presentations, Problem Solving, Risk Management, Stress Testing
#4	A	Information Security, Information Systems, Presentations, Risk Management, Software Engineering, Troubleshooting, Unix
#5	B	Information Systems, Leadership, Multitasking, Risk Management, Software Development, Systems Development, Troubleshooting, Basel, Derivatives, Risk Management, Stress Testing, Systems Development, Troubleshooting
#6	B	Leadership, Systems Development, Troubleshooting
#7	B	

Table 1 (continued)

#8	B	Writing, Multitasking, Problem Solving, Troubleshooting, Information Systems, Management, Information Security, Risk Management, Software Engineering
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Panel 3. This exhibit reports log-odds ratios for word-topic associations obtained by running STM on the job skill lists in Exhibit 2. Higher (lower) values represent stronger (weaker) associations. For instance, 'Basel' is strongly associated with Topic 3.

Word	Topic 1	Topic 2	Topic 3
Basel	-1000.0	-1000.0	-2.3
CCAR	-106.1	-51.0	-2.3
Communication	-77.7	-3.0	-2.9
Derivatives	-125.7	-81.7	-2.3
Economics	-106.1	-51.0	-2.3
Information Security	-2.1	-1000.0	-1000.0
Information System	-1.7	-52.4	-65.9
Leadership	-28.2	-2.3	-43.8
Management	-2.8	-2.3	-93.7
Multitask	-38.0	-1.6	-15.2
Presentations	-2.9	-1.8	-16.7
Problem Solving	-2.6	-36.5	-3.2
Risk Management	-1.8	-2.0	-1.9
Software Engineering	-2.1	-118.4	-152.7
Stress Testing	-106.1	-63.7	-1.9
Systems Development	-22.0	-18.8	-2.3
Teamwork	-79.9	-2.2	-3.1
Troubleshooting	-1.5	-3.2	-3.2
Writing	-1000.0	-2.3	-1000.0

Panel 4. Top FREX words. This exhibit reports the top five FREX (frequent and exclusive) words for each topic, obtained from running STM on the job skill lists in Exhibit 2.

Topic 1 (Technology)	Topic 2 (General Business)	Topic 3 (Capital Reg.)
Information System	Multitask	Stress Test
Troubleshoot	Presentations	Basel
Information Security	Write	CCAR
Software Engineering	Teamwork	Derivatives
Risk Management	Leadership	Economics

Panel 5. This exhibit reports topic proportion scores for each job skill list in Exhibit 2 as calculated by STM. These sum to one within each row, and we interpret them as the proportion of each job's skills that relate to each topic.

Job	Bank	Topic 1 (Technology)	Topic 2 (General Business)	Topic 3 (Capital Reg.)
#1	A	0.04	0.09	0.87
#2	A	0.12	0.79	0.09
#3	A	0.10	0.31	0.59
#4	A	0.81	0.13	0.06
#5	B	0.34	0.42	0.24
#6	B	0.09	0.11	0.81
#7	B	0.08	0.84	0.08
#8	B	0.86	0.10	0.05

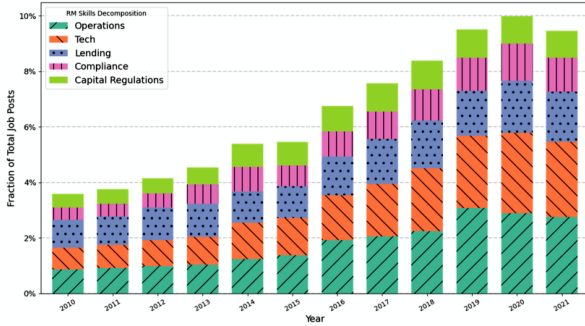
Table 1: Hypothetical Example to Illustrate Construction of RM-Skill

5.3 Figure 2: Demand for RM Skills Decomposition; Figure 3: Demand for RM Skills by Occupations

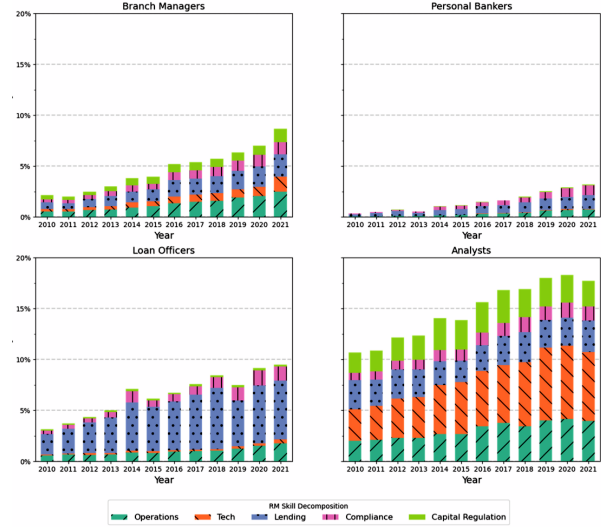
Read: Figure 2 decomposes the rise in RM Skills into Operations, Technology, Lending, Compliance, and Capital Regulation. Figure 3 shows the rise across branch managers, personal bankers, loan officers, and

analysts.

Economic message. Risk-management skills spread across ordinary banking functions rather than remaining inside a narrow risk department.



(a) Figure 2: Demand for RM Skills Decomposition



(b) Figure 3: Demand for RM Skills by Occupations

5.4 Table 3: Bank Characteristics and Labor Demand for RM Jobs; Table 4: Bank Characteristics and Labor Demand for RM Skills

Read: Table 3 links broad RM Jobs to risk and bank characteristics. Table 4 repeats the analysis using RM Skills. Poor performance and volatility matter for some groups, while large-bank status and deposit funding point to distorted incentives.

Economic message. The tables reveal competing forces: realized risk can raise RM demand, but safety-net and large-bank distortions weaken incentives for broad RM investment.

Table 3
Bank characteristics and labor demand for RM jobs.

	RM Jobs _{i,t}				F-test
	All Banks		Small Banks	Large Banks	
	(1)	(2)	(3)	(4)	
Volatility of ROA _{i,t-1}	1.421 (1.193)	1.542 (1.202)	1.833 (1.310)	-3.057 (2.338)	3.944** (0.048)
ROA _{i,t-1}	-2.404*** (0.747)	-2.324*** (0.743)	-2.299*** (0.816)	-4.544** (1.874)	0.819 (0.366)
Deposits /Assets _{i,t-1}	-0.063 (0.065)	-0.060 (0.064)	-0.056 (0.087)	-0.099 (0.091)	0.098 (0.755)
Cap Ratio _{i,t-1}	-0.242 (0.167)	-0.228 (0.166)	-0.259 (0.185)	0.070 (0.318)	1.127 (0.289)
Loans & Commit /Assets _{i,t-1}	-0.014 (0.035)	-0.025 (0.035)	-0.040 (0.053)	0.035 (0.042)	1.225 (0.269)
Complexity _{i,t-1}	1.231 (0.887)	0.194 (1.060)	0.280 (1.171)	0.464 (2.565)	0.001 (0.982)
Stress Tested _{i,t-1}	2.842 (2.412)	4.866** (2.428)	Absorbed (3.367)	2.277 (2.538)	Absorbed (0.001)
Large Banks _{i,t-1}	-6.253** (2.777)	-7.601** (3.367)	Absorbed	Absorbed	Absorbed
Year FE	Yes	Yes	Yes	Yes	
Size Controls	No	Yes	Yes	Yes	
Observations	3706	3706	3288	418	
Adjusted R ²	0.061	0.066	0.063	0.120	

This table reports results from Eq. (1) for all banks (columns 1 & 2), small banks

(a) Table 3: Bank characteristics and RM Jobs

Table 4
Bank characteristics and labor demand for RM skills.

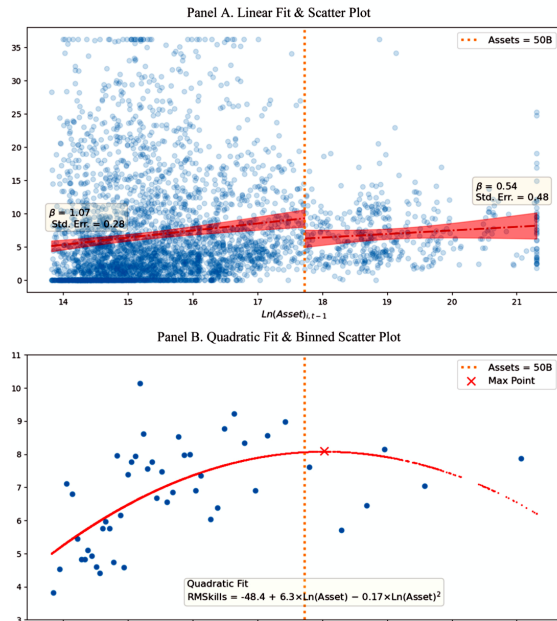
	RM Skills				F-test
	All Banks		Small Banks	Large Banks	
	(1)	(2)	(3)	(4)	
Volatility of ROA _{i,t-1}	0.643* (0.355)	0.694* (0.359)	0.776** (0.391)	-0.239 (0.570)	2.776* (0.096)
ROA _{i,t-1}	-0.737*** (0.244)	-0.708*** (0.244)	-0.680** (0.268)	-0.857* (0.480)	0.073 (0.787)
Deposits /Assets _{i,t-1}	-0.043** (0.021)	-0.040* (0.021)	-0.022 (0.028)	-0.084*** (0.023)	3.125* (0.078)
Cap Ratio _{i,t-1}	-0.090 (0.055)	-0.084 (0.055)	-0.089 (0.061)	0.169* (0.085)	6.842*** (0.009)
Loans & Commit /Assets _{i,t-1}	-0.009 (0.011)	-0.012 (0.011)	-0.015 (0.017)	0.003 (0.009)	0.832 (0.362)
Complexity _{i,t-1}	0.289 (0.274)	-0.072 (0.325)	0.037 (0.370)	-0.592 (0.406)	1.310 (0.253)
Stress Tested _{i,t-1}	0.509 (0.607)	0.959 (0.594)	Absorbed	1.299* (0.689)	Absorbed
Large Banks _{i,t-1}	-2.002** (0.799)	-2.632*** (0.997)	Absorbed	Absorbed	Absorbed
Year FE	Yes	Yes	Yes	Yes	
Size Controls	No	Yes	Yes	Yes	
Observations	3706	3706	3288	418	
Adjusted R ²	0.060	0.064	0.059	0.351	

(b) Table 4: Bank characteristics and RM Skills

5.5 Figure 4: Demand for RM Skills by Size; Table 5: Large Banks—Risk and Other Factors

Read: Figure 4 shows that RM Skills rise with size below the \$50B threshold but flatten or bend downward among the largest banks. Table 5 adds governance and market-risk variables for large banks.

Economic message. Size creates both complexity and safety-net distortions. Large banks need RM skill, but their incentives to invest broadly may be blunted.



(a) Figure 4: Demand for RM Skills by Size

Table 5 Large banks: risk and other factors.							
	RM Jobs				RM Skills		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Volatility of ROA _{i,t-1}	-2.961 (2.127)	-0.395 (0.514)	-0.451 (0.660)	-0.603 (0.600)	-0.448 (0.665)	-0.601 (0.602)	-0.518 (0.681)
ROA _{i,t-1}	-3.265* (1.846)	-0.666 (0.498)	-0.860 (0.528)	-0.634 (0.466)	-0.860 (0.529)	-0.631 (0.467)	-0.732 (0.523)
Deposits /Assets _{i,t-1}	-0.106 (0.076)	-0.087*** (0.021)	-0.078*** (0.025)	-0.086*** (0.022)	-0.078*** (0.026)	-0.085*** (0.024)	-0.087*** (0.025)
Cap Ratio _{i,t-1}	-0.036 (0.327)	0.139* (0.075)	0.150* (0.092)	0.121 (0.092)	0.159* (0.091)	0.128 (0.091)	0.133** (0.087)
Loans & Commit /Assets _{i,t-1}	0.044 (0.049)	0.005 (0.007)	0.002 (0.009)	0.004 (0.007)	0.002 (0.009)	0.004 (0.007)	0.005 (0.009)
Complexity _{i,t-1}	0.880 (2.544)	-0.451 (0.324)	-0.599 (0.418)	-0.438 (0.330)	-0.600 (0.418)	-0.437 (0.331)	-0.617 (0.406)
Stress Tested _{i,t-1}	3.032 (2.497)	1.600** (0.641)	1.149 (0.726)	1.315** (0.633)	1.151 (0.726)	1.313** (0.635)	1.217* (0.723)
Expected Shortfall _{i,t-1}			0.026 (0.059)	-0.003 (0.057)			
Beta _{i,t-1}					0.665 (1.713)	-0.027 (1.654)	
Volatility _{i,t-1}							-0.052 (0.041)
Long Term Investors _{i,t-1}	0.412** (0.204)	0.136*** (0.037)		0.151*** (0.037)			-0.050 (0.049)
Delta _{i,t-1}	1.104** (0.524)	-0.029 (0.127)		-0.038 (0.126)			-0.059 (0.130)
Vega _{i,t-1}	-11.485** (4.804)	-1.007 (0.727)		-0.994 (0.732)			-1.004 (0.756)
E-index _{i,t-1}	0.577 (1.342)	-0.069 (0.191)		-0.044 (0.207)			-0.039 (0.216)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	418	418	398	398	398	398	398
Adjusted R ²	0.180	0.438	0.349	0.442	0.349	0.442	0.355

This table reports estimates from Eq. (1) with three additional governance factors (Long-term Investors, Delta, Vega) and market-based risk metrics (Expected Shortfall, Beta, and Volatility). RM Skills are the sum of topic scores from risk management related topics scaled by total job posts for a given bank-year. The sample includes all banks with total assets no less than \$50B. Size controls include log(Assets) and its square. Standard errors are clustered at the bank level. Please see Appendix Table 7 for

(b) Table 5: Large banks: risk and other factors

5.6 Table 6: Decompose RM Skills; Figure 5: Ever Stress-Tested vs. Never Stress-Tested Banks

Read: Table 6 shows that deposits negatively predict several RM skill categories for large banks, while capital ratios and stress-test exposure matter for capital regulation. Figure 5 shows similar upward trends but different composition across ever-tested and never-tested banks.

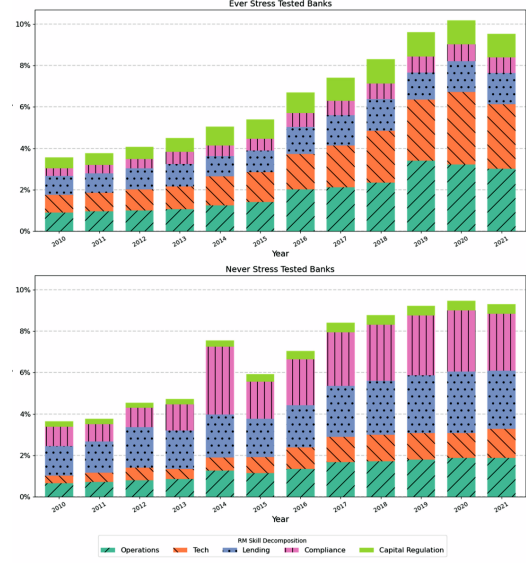
Economic message. Stress-tested banks' RM demand is tilted toward technology and capital regulation, whereas other banks' RM demand contains more broad compliance and operations content.

Table 6

Decompose RM Skills.

	Operations _{it}	Tech _{it}	Lending _{it}	Compliance _{it}	Cap Reg _{it}
Panel A: Large Banks					
(1)	(2)	(3)	(4)	(5)	
Volatility of	0.003	0.203	-0.307*	-0.175*	0.217
ROA _{it-1}	(0.157)	(0.182)	(0.167)	(0.099)	(0.132)
ROA _{it-1}	-0.179	0.046	-0.396**	-0.240***	0.055
	(0.136)	(0.170)	(0.102)	(0.084)	(0.119)
Deposits	-0.016***	-0.010***	-0.004	-0.006**	-0.022***
/Assets _{it-1}	(0.006)	(0.008)	(0.005)	(0.003)	(0.005)
Cap Ratio _{it-1}	0.045*	0.083***	-0.039*	0.002	0.062***
	(0.023)	(0.026)	(0.020)	(0.013)	(0.021)
Loans & Commit	0.002	0.003	0.001	0.001	-0.005*
/Assets _{it-1}	(0.002)	(0.004)	(0.002)	(0.001)	(0.003)
Complexity _{it-1}	-0.204	-0.173	-0.010	-0.038	-0.120*
	(0.124)	(0.153)	(0.096)	(0.060)	(0.070)
Stress	0.292	0.165	0.211	-0.015	0.339**
Tested _{it-1}	(0.180)	(0.208)	(0.154)	(0.097)	(0.132)
Year FE	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes
Observations	418	418	418	418	418
Adjusted R ²	0.291	0.396	0.161	0.188	0.586
Panel B: Small Banks					
Volatility of	0.179**	0.136**	0.204	0.219	0.053***
ROA _{it-1}	(0.076)	(0.081)	(0.138)	(0.166)	(0.020)
ROA _{it-1}	-0.137***	-0.066	-0.301***	-0.137	-0.049***
	(0.052)	(0.043)	(0.083)	(0.108)	(0.016)
Deposits	-0.005	-0.006	-0.005	-0.002	-0.006*
/Assets _{it-1}	(0.005)	(0.004)	(0.009)	(0.012)	(0.003)
Cap Ratio _{it-1}	-0.012	-0.032*	-0.033	-0.003	-0.003
	(0.012)	(0.011)	(0.016)	(0.022)	(0.004)
Loans & Commit	-0.002	-0.001	0.001	-0.008	-0.001
/Assets _{it-1}	(0.003)	(0.002)	(0.005)	(0.007)	(0.001)
Complexity _{it-1}	0.002	0.042	-0.057	0.073	-0.011
	(0.068)	(0.058)	(0.117)	(0.155)	(0.020)
Year FE	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes
Observations	3288	3288	3288	3288	3288
Adjusted R ²	0.067	0.051	0.055	0.042	0.098

(a) Table 6: Decompose RM Skills

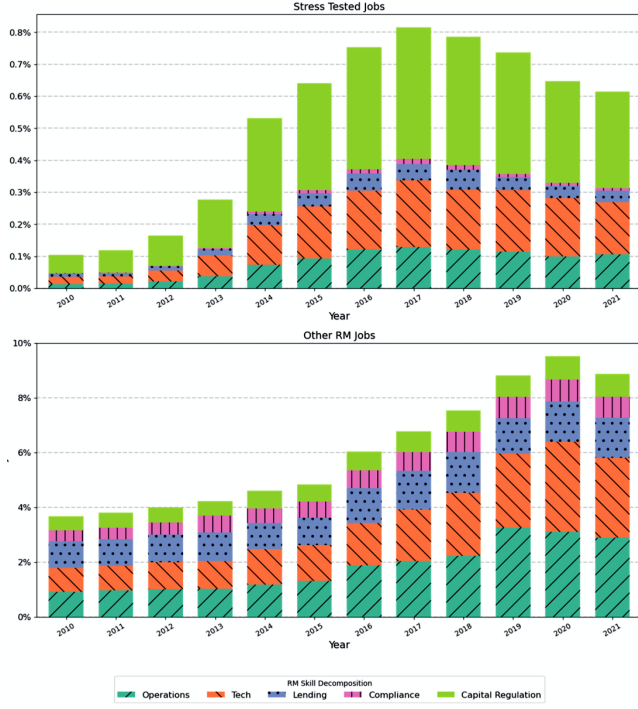


(b) Figure 5: Ever Stress-Tested vs. Never Stress-Tested Banks

5.7 Figure 6: Stress Test Jobs vs. Other RM Jobs; Table 7: Summary Statistics

Read: Figure 6 separates stress-test jobs from other RM jobs. Table 7 shows that stress-test jobs are only about 1.37% of job posts, while other RM jobs are about 21.38%. Stress-test skills are about 10% of total RM skills.

Economic message. Stress-test human capital is important but narrow. The key policy question is whether it spills over to the rest of risk management.



(a) Figure 6: Stress Test Jobs vs. Other RM Jobs

Table 7

Summary statistics.

	N	Mean	Std	p25	p50	p75
Panel A. Stress Test Jobs						
Stress Test Jobs _{it}	398	1.37	1.59	0.34	0.76	1.80
Higher Edu _{it}	398	0.31	0.37	0.06	0.18	0.41
Low Edu _{it}	398	1.06	1.32	0.22	0.55	1.34
RM Skills _{it}	398	0.68	0.81	0.15	0.38	0.88
Operations _{it}	398	0.10	0.13	0.02	0.05	0.13
Tech _{it}	398	0.17	0.22	0.04	0.09	0.21
Lending _{it}	398	0.04	0.05	0.01	0.03	0.06
Compliance _{it}	398	0.01	0.01	0.00	0.01	0.01
Capital Regulation _{it}	398	0.36	0.43	0.08	0.19	0.47
Panel B. Other RM Jobs						
Other RM Jobs _{it}	398	21.38	16.45	9.98	16.93	26.48
Higher Edu _{it}	398	1.06	1.05	0.35	0.73	1.38
Low Edu _{it}	398	20.32	16.28	9.11	15.84	25.17
RM Skills _{it}	398	6.31	3.60	3.43	5.77	8.97
Operations _{it}	398	1.61	1.10	0.82	1.39	2.12
Tech _{it}	398	1.72	1.50	0.67	1.28	2.16
Lending _{it}	398	1.37	0.91	0.72	1.13	1.71
Compliance _{it}	398	0.76	0.56	0.36	0.56	0.99
Capital Regulation _{it}	398	0.86	0.79	0.33	0.59	1.09
Panel C. Bank Characteristics						
Stress Tested _{it-1}	398	0.78	0.41	1.00	1.00	1.00
Stress Tested _{it}	398	0.72	0.45	0.00	1.00	1.00
Volatility of ROA _{it-1}	398	0.26	0.48	0.05	0.11	0.24
ROA _{it-1}	398	0.77	0.86	0.51	0.81	1.10
Loans/Assets _{it-1}	398	89.54	52.32	66.58	86.67	102.80
Deposits/Assets _{it-1}	398	59.41	21.35	49.91	68.24	75.03
Capital Ratio _{it-1}	398	13.75	3.88	11.42	12.75	14.68
Complexity _{it-1}	398	5.32	1.40	4.29	4.82	6.54
Panel D. Stress Test Results						
Tier1 Cap Exposure _{it-1}	215	2.28	2.49	0.65	1.53	3.44
TRBC Exposure _{it-1}	215	2.28	2.60	0.54	1.50	3.59
Tier1 Lev Exposure _{it-1}	215	1.18	1.19	0.45	1.06	2.01
CCAR Failure Score _{it-1}	215	0.21	0.53	0.00	0.00	0.00

This table lists the summary statistics for labor demand variables (Panel A and B), bank characteristics (Panel C) and stress test results (Panel D). RM Jobs are categorized as *Stress Test Jobs* if it requires "Stress Testing" or "CCAR Regulatory

(b) Table 7: Summary Statistics

5.8 Figure 7: Dynamic Difference-in-Difference around Stress Test Initiation

Read: The treatment banks show a sharp rise in stress-test RM skills around 2014–2015, but other RM jobs do not show a comparable rise.

Economic message. Stress-test entry causes a burst of test-specific human-capital investment, not a broad increase in all RM hiring.

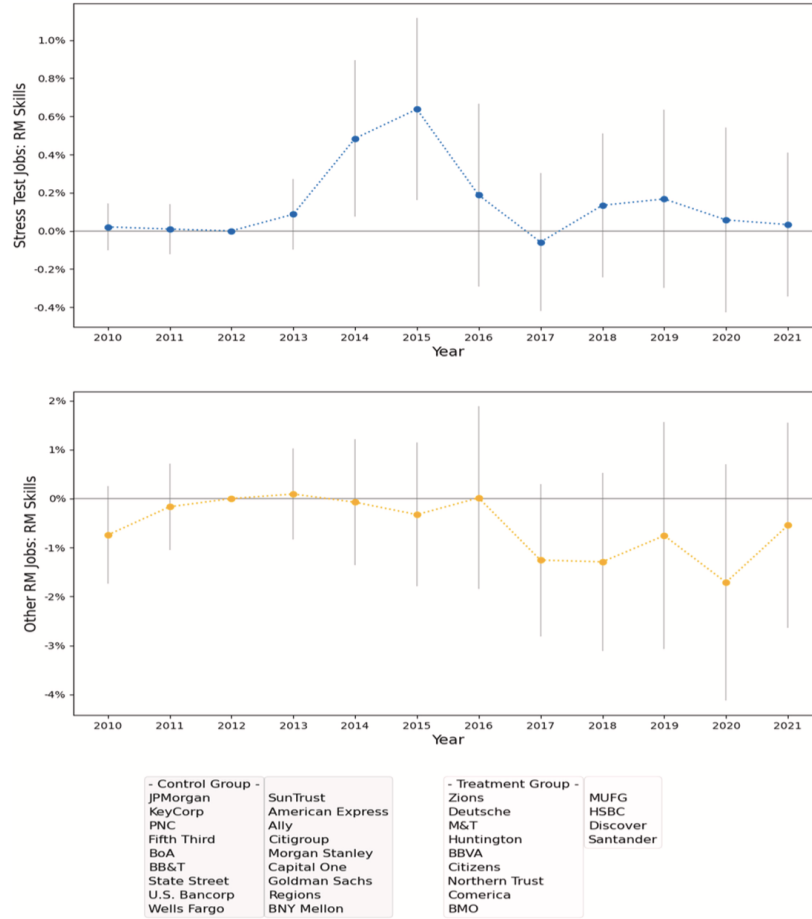


Fig. 7. Dynamic Difference-in-Difference around Stress Test Initiation
This figure plots the difference in labor demand for stress-test jobs (upper panel) and other RM jobs (lower panel) between the treatment and control group. The treatment group includes a set of banks that join the test in 2014 (or 2015) for the first time, whereas the control group includes banks that started their tests in 2011. We regress the *RM skills* for stress-test jobs (or other RM jobs) on an interaction term between Treated and Year with bank and year fixed effects and standard errors clustered by bank. The year 2012 is the omitted year and the coefficients of the interaction term are plotted with 90 % confidence intervals.

Figure 7: Dynamic Difference-in-Difference around Stress Test Initiation

5.9 Tables 8–9: Stress Tests and Stress-Test Job Skills

Read: Table 8 shows that an upcoming stress test raises Stress Test Jobs by 0.450 percentage points and especially raises highly educated stress-test jobs. Table 9 shows that the increase is concentrated in Operations, Technology, Compliance, and Capital Regulation skills.

Economic message. Stress tests produce specialized regulatory and technical hiring rather than broad RM hiring.

5.10 Tables 10–11: Stress-Test Failure, Exposure, and Skill Demand

Read: Table 10 shows that CCAR failure scores increase stress-test job demand but not other RM jobs. Table 11 shows that failure increases stress-test RM skills, especially Operations, Technology, and Capital Regulation.

Table 8
Stress tests and labor demand for stress test jobs.

	Stress Test Jobs _{it}			Other RM Jobs _{it}		
	All (1)	Higher Edu (2)	Low Edu (3)	All (4)	Higher Edu (5)	Low Edu (6)
Stress	0.450*** (0.152)	0.158*** (0.052)	0.285** (0.121)	-0.500 (1.731)	-0.003 (1.731)	-0.524 (1.689)
Stress _{it-1}	0.173 (0.154)	-0.002 (0.048)	-0.180 (0.131)	-3.856*** (1.874)	-0.058 (0.992)	-3.759** (1.875)
Volatility	0.441** (0.171)	0.344*** (0.042)	0.433 (0.162)	-0.238*** (0.097)	0.871 (1.530)	0.471 (1.520)
if ROA _{it-1}	0.265*** (0.095)	0.220*** (0.032)	-0.915 (1.080)	-0.212** (1.280)	-0.499 (1.364)	-0.138 (1.364)
Deposits	0.018* (0.010)	-0.003 (0.010)	0.020* (0.010)	-0.153 (0.085)	-0.015*** (0.010)	-0.138 (0.101)
/Assets _{it-1}	0.119*** (0.023)	0.037* (0.010)	0.099*** (0.022)	0.223 (0.345)	-0.011 (0.030)	0.236 (0.337)
Loan & Commit	-0.002 (0.001)	-0.002 (0.001)	-0.002 (0.001)	-0.002 (0.041)	-0.002 (0.007)	-0.006 (0.039)
Complexity _{it-1}	0.276** (0.147)	0.323** (0.042)	0.326 (0.124)	3.526 (3.273)	0.247 (3.238)	3.303 (3.268)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	398	398	398	398	398	398
Adjusted R ²	0.721	0.570	0.690	0.576	0.618	0.577

This table shows the estimates of Eq. (2) with *Stress Test Jobs* and *Other RM Jobs* as the outcome variables. *Stress Test Jobs* are RM jobs that require "Stress Testing" or "CCAR Regulatory Rules". A job is considered as *Higher Edu* if it requires a master or doctoral degree and *Low Edu* otherwise. Size control includes log(Assets) and its square. Standard errors are clustered at the bank level. Please see Appendix Table 3 for detailed variable definitions.

Table 10
Exposure to stress tests and labor demand for stress test jobs.

	Stress Test Jobs _{it}			Other RM Jobs _{it}		
	All (1)	Higher Edu (2)	Low Edu (3)	All (4)	Higher Edu (5)	Low Edu (6)
CCAR Failure	0.216** (0.127)	0.068** (0.029)	0.237* (0.120)	-0.130 (1.965)	0.193 (0.119)	-0.323 (1.937)
Tier1 Cap	0.025 (0.196)	-0.008 (0.041)	0.018 (0.171)	2.352 (2.281)	-0.106 (0.123)	2.459 (2.242)
Exposure _{it-1}	0.091 (0.166)	0.024 (0.035)	0.070 (0.145)	-1.482 (1.779)	0.178 (0.114)	-1.662 (1.780)
Tier1 Lev	-0.035 (0.123)	0.037* (0.021)	-0.062 (0.106)	-1.105 (1.816)	-0.011 (0.066)	-1.093 (1.799)
Volatility	0.496** (0.203)	-0.011 (0.057)	0.512*** (0.187)	-2.566 (2.822)	-0.240 (0.219)	-2.566 (2.713)
if ROA _{it-1}	0.413*** (0.143)	0.028 (0.036)	0.388*** (0.126)	-1.883 (2.245)	-1.883 (0.148)	-1.861 (2.198)
Deposits	-0.019 (0.011)	-0.013 (0.003)	-0.013 (0.010)	0.135 (0.138)	0.005 (0.011)	0.131 (0.133)
Cap Ratio _{it-1}	0.027 (0.048)	-0.006 (0.039)	0.036 (0.041)	0.188 (0.962)	0.065 (0.047)	0.123 (0.963)
Loan & Commit	0.019** (0.009)	0.003 (0.003)	0.015** (0.007)	0.133 (0.134)	-0.010 (0.020)	0.143 (0.131)
Complexity _{it-1}	0.351* (0.207)	0.311 (0.047)	0.420 (0.191)	4.996 (3.501)	0.024 (0.210)	4.970 (3.373)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	215	215	215	215	215	215
Adjusted R ²	0.788	0.606	0.783	0.706	0.685	0.708

This table shows the estimates of Eq. (3) using *Stress Test Jobs* and *Other RM Jobs* as dependent variables. *Stress Test Jobs* are RM jobs that require "Stress Testing" or "CCAR Regulatory Rules". Size control includes log(Assets) and its square. Standard errors are clustered at the bank level. Please see Appendix Table 3 for detailed variable definitions.

6 Short summary

- RM labor demand rises sharply from 2010 to 2021.
- The NLP-based RM Skills measure shows that technical and regulatory RM skills are a smaller subset of broad RM hiring.
- Stress tests raise demand for stress-test-specific jobs, especially highly educated jobs.
- The increase is strongest in Technology and Capital Regulation.
- Stress tests do not increase broad, non-test RM hiring.
- Stress-test failures increase stress-test-specific hiring and skill demand.

Table 9
Stress test and stress test jobs RM skills decomposition.

	Panel A. Stress Test Jobs					
	RM Skills _{it} (1)	Operations _{it} (2)	Tech _{it} (3)	Lending _{it} (4)	Compliance _{it} (5)	Cap Reg _{it} (6)
Stress	0.239*** (0.091)	0.032*** (0.011)	0.058** (0.023)	0.008 (0.008)	0.003* (0.002)	0.137** (0.057)
Stress _{it-1}	0.086 (0.080)	0.032 (0.023)	0.057 (0.023)	0.007 (0.006)	0.000 (0.002)	0.037 (0.044)
Volatility	0.193** (0.079)	0.030** (0.015)	0.052** (0.025)	0.009 (0.006)	0.003 (0.002)	0.095** (0.038)
ROA _{it-1}	0.151*** (0.051)	0.013 (0.008)	0.050** (0.018)	0.003 (0.003)	0.001 (0.001)	0.085*** (0.026)
Deposits	0.006 (0.005)	0.002** (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.000)	0.002 (0.003)
/Assets _{it-1}	0.055*** (0.014)	0.008*** (0.002)	0.016*** (0.005)	0.002*** (0.001)	0.001*** (0.000)	0.028*** (0.009)
Loan & Commit	-0.000 (0.002)	-0.000 (0.000)	-0.000 (0.001)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.001)
Complexity _{it-1}	0.174** (0.082)	0.034** (0.015)	0.046** (0.024)	0.013** (0.005)	0.003** (0.001)	0.078 (0.055)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	398	398	398	398	398	398
Adjusted R ²	0.703	0.685	0.636	0.654	0.596	0.694

	Panel B. Other RM Jobs					
	RM Skills _{it} (1)	Operations _{it} (2)	Tech _{it} (3)	Lending _{it} (4)	Compliance _{it} (5)	Cap Reg _{it} (6)
Stress	0.235 (0.337)	0.089 (0.129)	0.244 (0.156)	0.118 (0.128)	-0.056 (0.063)	0.030 (0.062)
Stress _{it-1}	-0.945** (0.251)	-0.193* (0.099)	-0.195 (0.125)	-0.223* (0.122)	-0.109 (0.048)	-0.244*** (0.068)
Volatility	-0.418 (0.266)	-0.175** (0.078)	-0.214 (0.140)	0.029 (0.090)	-0.089* (0.046)	0.017 (0.060)
if ROA _{it-1}	-0.264*** (0.083)	-0.264*** (0.095)	-0.332*** (0.160)	-0.132*** (0.078)	-0.060 (0.053)	-0.077 (0.040)
Deposits	-0.070*** (0.021)	-0.021** (0.009)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.006)	-0.007 (0.004)
/Assets _{it-1}	0.035 (0.007)	0.026 (0.007)	-0.020 (0.007)	-0.017 (0.012)	0.049*** (0.012)	-0.018 (0.003)
Loan & Commit	-0.004 (0.010)	-0.003 (0.003)	0.004 (0.003)	-0.003 (0.003)	-0.003 (0.002)	0.000 (0.002)
Complexity _{it-1}	0.317 (0.687)	-0.014 (0.287)	0.170 (0.274)	0.095 (0.155)	-0.033 (0.093)	0.152 (0.127)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	398	398	398	398	398	398
Adjusted R ²	0.657	0.592	0.613	0.635	0.591	0.782

This table shows the estimates of Eq. (2) when RM Skills and the subcategories of RM skills of stress test jobs are the outcome variables in Panel A, and other RM jobs in Panel B. *Stress Test Jobs* are RM jobs that require "Stress Testing" or "CCAR Regulatory Rules". Size control includes log(Assets) and its square. Standard errors are clustered at the bank level. Please see Appendix Table 3 for detailed variable definitions.

- The policy implication is mixed: stress tests build specialized capacity but do not clearly transform broad risk-management culture.

7 Expanded conclusion

7.1 Core conclusion

Stress testing affects bank human-capital investment, but the effect is narrow. Banks hire people with stress-test-specific skills, especially after upcoming tests or test failures.

7.2 Economic intuition

Stress tests require banks to produce models, data pipelines, capital plans, and regulatory documentation. This creates demand for technical and capital-regulation human capital. But because the regulatory task is narrow, hiring may remain compliance-oriented.

7.3 Policy interpretation

The evidence supports both an optimistic and pessimistic view. Stress tests force banks to build capacity they may otherwise underinvest in, but the capacity is concentrated in the parts of risk management directly evaluated by supervisors.

7.4 Limitations and future research

The paper measures job demand, not actual risk reduction. Future work could test whether stress-test-driven hiring improves loan screening, portfolio risk, crisis performance, or internal risk culture.

7.5 Takeaway

Regulation can reshape bank human capital. To improve resilience, stress tests may need to encourage spillovers from test-specific expertise into broader day-to-day risk management.